

Reference Runbook: Cross-Tower RCA Inference

RCA should begin from the user-facing incident and then explain supporting cross-tower evidence.

Known causes and recommended actions:

Storage latency caused downstream application timeouts: database read/write latency rises before application timeout errors. Recommended action: validate volume health, IOPS throttling, queue depth, and fail over or rebalance storage.

Application deployment or code regression: deployment errors, new 5xx patterns, or configuration changes precede the user impact. Recommended action: compare release timeline, roll back, and inspect changed dependency handling.

Network degradation between service dependencies: packet loss, DNS failures, connection resets, or load balancer latency precede dependency failures. Recommended action: validate route, DNS, target health, packet drops, and regional blast radius.

Compute resource pressure: CPU, memory, pod restarts, or node health anomalies align with the incident window. Recommended action: inspect node pressure, scaling events, crash loops, and resource limits.

Dependency outage: one downstream service error dominates multiple upstream application symptoms. Recommended action: isolate dependency calls, check health endpoints, and activate fallback or circuit breaker.

RCA generation rules:

Treat application errors and latency as symptoms unless deployment evidence is the strongest leading indicator.

Prefer causes whose evidence starts before the user-facing symptom.

Always include confidence, supporting evidence, alternative hypotheses, and reasons alternatives ranked lower.

Include recommended validation steps from this runbook.

Do not claim causation from correlation alone. Explain uncertainty and recommend validation steps.